

BASIC HUMAN TASTES

Four modules for VCV Rack · manuals

ERNEST	Percussion oscillator with six ways to bend its pitch
MARCEL	8-bit looper that slowly forgets
JULES	Six linked slopes: envelopes, LFOs, oscillators, chords
ODETTE	Stereo repeater, from plucked strings to forty-second loops

Four voices for playing live and building generative patches: a percussion oscillator, a looper that forgets, six linked slopes and a stereo repeater.

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ERNEST

Percussion oscillator with six ways to bend its pitch

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Ernest is a single percussion voice built around an oscillator whose pitch can be bent six different ways. Hit it with a trigger and you get kicks, toms, snares, zaps and metallic pings. Leave the trigger unpatched and it plays continuously, which is handy for tuning.

Beyond the classic drum-machine voice it has a few modular extras: a pitch input, CV over the modulation, a modulation output and a free-running mode.

AT A GLANCE

- Sine or triangle oscillator, 20 Hz to 12 kHz, with a 1 V/oct input
- Six pitch modulation shapes, from a falling saw to a pitch envelope
- Modulation from 0.1 Hz to 5 kHz: slow wobbles turn into new tone colours
- One-knob decay envelope, bass boost with drive, ring modulation
- 12 HP

CONTROLS

Screen	The modulation shape and the base pitch.
PITCH	Base pitch, from 20 Hz to 12 kHz. It starts at 55 Hz.
WAVE	Lit: triangle, slightly brighter. Off: sine.
RETRIG	Lit (default): every hit restarts the modulation, so every hit sounds the same. Off: the modulation keeps running and each hit lands somewhere else in its cycle.
Shape selector	The six icons around this knob: saw down (the pitch falls, again and again), square (two pitches alternate), triangle (rises and falls), random (a new random step each cycle), noise (bursts of noise in the pitch, for snares) and envelope (one sweep down after each hit, for kicks and toms).
DEPTH	How far the pitch moves, from -100 % to +100 %. Centre: no modulation. Negative values flip the direction. Gentle near the centre, up to ± 8 octaves at the ends.
RATE	Modulation speed, from 0.1 Hz to 5 kHz. With the envelope shape, how fast the sweep falls back. Above about 20 Hz the modulation is heard as a new tone colour rather than movement.
DECAY	Length of the volume envelope, from 3 ms to 3 s.
LEVEL	Output level.
BASS	Boosts the low end, then drives the sound into saturation as you turn it up.
RING	Lit: the oscillator is multiplied by the signal at RING IN. Nothing comes out unless both are sounding.

INPUTS

TRIG	Plays a hit: restarts the oscillator, both envelopes and, with RETRIG lit, the modulation. The small light flashes on each hit. Unpatched, Ernest plays continuously.
V/OCT	Added to PITCH, 1 volt per octave.
DEPTH	Added to the DEPTH knob; ± 5 V covers the whole range.
RATE	Added to the RATE knob; 1 V moves it a tenth of its travel.
RING IN	The signal used by ring modulation (± 5 V).

OUTPUTS

MOD	The pitch modulation itself, after DEPTH: 5 V for 8 octaves. Use it to move other modules in step with each hit.
OUT	Audio, ± 5 V at full LEVEL.

HOW IT WORKS

Which way the pitch moves

Saw down, square, triangle and envelope start from PITCH and move up from it (or down, with negative DEPTH), then come back. Random and noise move both ways around PITCH.

The envelope jumps to full on every hit and falls back at a speed set by RATE. Noise is strongest at the start of each modulation cycle and fades towards its end, which gives snares their crack.

PATCH IDEAS

1. Analog kick

Ernest's starting settings: envelope shape, DEPTH 50 %, RATE at the centre, PITCH 55 Hz, DECAY around two o'clock. Patch a clock into TRIG.

2. Snare

Noise shape, WAVE lit, PITCH around 200 Hz, DEPTH 40 %, RATE far right, DECAY short, a little BASS.

3. Tuned toms

Envelope shape, DEPTH 30 %, a short DECAY, and a pitch sequence into V/OCT.

4. Metallic ping

Two Ernests triggered together. Patch the first OUT into the second RING IN and light RING on the second. Raise RATE on either for inharmonic overtones.

5. Zap

Envelope shape, DEPTH at 100 %, RATE low for a slow sweep, DECAY short.

SPECIFICATIONS

Width	12 HP
Pitch	20 Hz to 12 kHz, plus 1 V/oct input
Modulation	0.1 Hz to 5 kHz, up to ± 8 octaves
Decay	3 ms to 3 s
Levels	Audio ± 5 V. MOD output 5 V per 8 octaves. Trigger threshold about 1 V.

MARCEL

8-bit looper that slowly forgets

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Marcel records into a small 8-bit memory read by a clock you can bend. Slow the clock and the loop gets longer, lower and grittier; speed it up and it gets short, clean and high.

And Marcel forgets. ERODE wears the loop down a little more on every pass and TONE darkens or thins it, until only a ghost of the sound is left. A ring at the top shows the memory as it fades.

AT A GLANCE

- 16,384 cells of 8-bit memory, clocked from 1 to 48 kHz: loops from 16 s down to 0.34 s
- Changing the speed bends the pitch of what is already recorded
- ERODE and TONE act on every pass, even on a frozen loop
- Reverse, reset, freeze, an end-of-loop trigger and a clock output
- 12 HP

CONTROLS

Ring	Each spoke is the loudness of one slice of the memory; the white dot is the play head. Gold while recording, cream while frozen.
SPEED	Memory clock, from 1 to 48 kHz. The loop lasts 16,384 clock ticks: 16 s fully left, 0.34 s fully right (1.6 s at the start). Turning it while sound is in the memory changes its pitch.
INPUT	Level of the incoming sound written into the memory.
FEEDBACK	How much of the loop is written back on each pass, up to 110 %. Above 100 % the loop grows until it saturates in 8 bits.
ERODE	How much the memory forgets on every pass: a little volume, less resolution (from 8 bits down to 4), slices that fade to silence and soft crackles. Fully right, a frozen loop has nearly gone after 4 passes; around 30 % it takes about a dozen.
PITCH	Amount of the PITCH input. Fully right, 1 V per octave on the clock; left of centre inverts it.
TONE	A filter inside the loop. Centre: no change. Left: each pass a little darker. Right: each pass a little thinner. Its cutoff follows SPEED.
FREEZE	Lit: nothing new is recorded. ERODE and TONE keep working, so a frozen loop still fades unless ERODE is at zero and TONE at the centre.

INPUTS

IN	Sound to record (± 5 V).
PITCH	Bends the memory clock, scaled by the PITCH knob.
ERODE	Added to the ERODE knob; 10 V = 100 %.
REVERSE	Plays and records backwards while the gate is high. The light next to it shows it.
RESET	A trigger sends the play head back to the start of the loop.
TRIG	Each trigger turns FREEZE on or off.
INPUT OPEN / CLOSE	Two level controls for the input. A rising CV (0 to 10 V) opens OPEN and closes CLOSE. Unpatched, they let everything through.
FEEDBACK OPEN / CLOSE	The same for the feedback.

OUTPUTS

END	A 10 V trigger each time the play head completes the loop, frozen or not. The light flashes.
CLOCK	Square wave at the memory clock divided by 32 (± 5 V): a tone that follows SPEED.
OUT	The memory, ± 5 V. Each value is held until the next clock tick, with no smoothing: that is where the grain comes from.

HOW IT WORKS

One cell per tick

On each clock tick Marcel reads one cell, then writes the input plus the feedback back into it and moves on. The loop therefore always lasts exactly 16,384 ticks, and recorded sound plays back faster or slower as the clock changes.

How forgetting works

Whatever comes round again passes through TONE and ERODE before it is written back. With ERODE above zero it loses a little volume and resolution, so quiet parts disappear first; now and then a slice fades to silence or a small crackle appears. The ring makes it visible: watch it hollow out.

PATCH IDEAS

1. A voice that crumbles

Microphone into IN, FEEDBACK 90 %. Record a phrase, press FREEZE, then raise ERODE to around 50 % and let it fall apart.

2. Varispeed tape

A slow triangle LFO into PITCH, PITCH knob around 30 %, FEEDBACK 70 %.

3. Back and forth

A slow square LFO into REVERSE.

4. Record one pass, play the next

END into TRIG: the loop freezes after one pass and opens again on the next.

5. Hear the loop length

END into the TRIG of an Ernest: one hit per pass.

SPECIFICATIONS

Width	12 HP
Memory	16,384 × 8 bits
Clock	1 to 48 kHz, plus CV
Loop length	0.34 to 16 s
Levels	Audio ±5 V. END 10 V, 1 ms. Gates and triggers about 1 V.

JULES

Six linked slopes: envelopes, LFOs, oscillators, chords

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Jules is six slope generators that share one set of controls. RATE sets how fast the first one goes; SPREAD sets how the other five relate to it: all the same, or 2, 3, 4, 5 and 6 times faster, or slower.

Depending on the range and the mode, the six slopes become cascading envelopes, polyrhythmic LFOs, six slew limiters, a chord of oscillators, plucked notes or FM voices. CHANCE adds controlled randomness to the way triggers travel across the six.

AT A GLANCE

- 6 slopes sharing RATE, SPREAD, RISE and CURVE
- CV range (envelopes and LFOs, 0 to 8 V) or AUDIO range (oscillators, ± 5 V)
- 12 modes on one selector, each one explained on the screen
- Triggers cascade from right to left, with a CHANCE control
- Six outputs, a mix and a 6-channel poly output
- 18 HP

CONTROLS

Screen	Seven bars (the six slopes and the mix), the mode name, and what TWEAK does in that mode.
RATE	Speed of slope 1. CV range: from about 2 minutes to 8 ms per cycle, 1 Hz at the centre. AUDIO range: about 12 Hz to 6 kHz, middle C at the centre.
SPREAD	Speed of slopes 2 to 6 compared with slope 1. Centre: all the same. Right: slope N goes up to N times faster (fully right $\times 2$, $\times 3$... $\times 6$, a harmonic series). Left: up to N times slower (an undertone series). Finer near the centre.
FM	Amount of the FM input. Right of centre: modulates the speed of all six slopes equally. Left of centre: modulates SPREAD, so slope 1 is untouched and the fundamental stays solid. Centre: off.
RISE	Share of each cycle spent rising, without changing its length. Left: instant rise, long fall (saw). Centre: triangle. Right: long rise, instant fall (ramp).
CURVE	Bends the slopes without changing their timing. Centre: straight lines. Right: exponential, then sine. Left: logarithmic, then square; fully left, RISE becomes the pulse width.
CV / AUDIO	Range switch. CV: slow, outputs 0 to 8 V. AUDIO: fast, outputs ± 5 V.
MODE	Six positions, each with a CV and an AUDIO version: see the mode table.
TWEAK	A second control for modes 2, 4 and 6. The small light shows when it is active; the screen says what it does.
CHANCE	Probability that a trigger passed along from the right reaches each slope. 100 % (default): every slope fires. 50 %: each one flips a coin on every trigger. Triggers patched directly always pass.

MODES

#	CV range	AUDIO range
1	ONE-SHOT. Each trigger draws one rise and fall. Triggers that arrive while a slope is moving are ignored.	IMPULSE. The same at audio rate: patch an audio-rate clock into TRIG for impulse-train oscillators.
2	RETRIGGER. Like ONE-SHOT, but TWEAK sets from when a moving slope accepts a new trigger: -5 V at any time, 0 V after its peak, $+5$ V only once finished. It then restarts from zero.	SUBHARMONIC. Slope 1 runs freely and fires the other five at the end of each of its cycles. TWEAK sets when they accept it. With SPREAD left: subharmonics, split tones and sync chaos.
3	GATE. Rises while the gate is high, holds at the top, falls when the gate drops.	PULSE. The same with an audio-rate clock: oscillators whose shape follows the clock's pulse width.
4	SUSTAIN. Rises, falls to a sustain level set by TWEAK (-5 V: 0 %, $+5$ V: 100 %), holds while the gate is high, then falls. With a steady gate, the six outputs follow TWEAK at six speeds: six slew limiters.	PLUCK. Six oscillators, each through its own lowpass gate. Triggers pluck notes, gates hold them. TWEAK sets the decay: about 300 ms at the centre, shorter to the right, longer to the left.
5	LOOP. Six free-running LFOs. Triggers reset them to the start of their cycle.	OSC. Six oscillators; triggers hard-sync them. SPREAD fully right gives a major chord, fully left a minor chord, across a few octaves.
6	BURST. Slopes rest until triggered, then run a burst of cycles: TWEAK -4 V: 1, 0 V: 6, $+5$ V: 36. Below -4 V triggers are ignored.	FM. Each oscillator gets its own sine modulator. TWEAK sets its ratio ($\times 0.5$ at -5 V, $\times 1$ at 0 V, $\times 2$ at $+5$ V), the FM knob the depth (right: equal, left: more on the higher slopes), the FM input scales it.

INPUTS

V/OCT	Added to RATE, 1 volt per octave: the speed doubles with each volt.
SPREAD · RISE · CURVE	Added to their knobs; ± 5 V covers the whole range.
FM	Frequency modulation source. In the AUDIO range a DC blocker keeps the pitch centred. In FM mode it becomes a depth control.
TWEAK	Added to the TWEAK knob, ± 5 V.
CHANCE	Added to the CHANCE knob; 10 V = 100 %.
TRIG 1 to 6	Trigger or gate for each slope, threshold about 1 V. Unpatched inputs take the signal from the right (see the trigger cascade).

OUTPUTS

1 to 6	Each slope: 0 to 8 V in the CV range, ± 5 V in the AUDIO range.
MIX	CV range: the highest of the six slopes, each divided by its number. AUDIO range: the six summed and gently limited, about ± 7.5 V.
POLY	The six slopes in one 6-channel cable.

HOW IT WORKS

The trigger cascade

Each TRIG input that has no cable receives the signal of the nearest patched input to its right. One cable into TRIG 6 therefore drives all six slopes; a second cable into TRIG 3 splits them into two groups (6, 5, 4 and 3, 2, 1).

CHANCE acts on that cascade only: every time a trigger travels to a slope through the cascade, the slope decides at random whether to take it. This is the quickest way to make six voices play a different combination on every beat while staying in time.

The MIX output

In the CV range, MIX outputs the highest of the six slopes, each divided by its number (slope 3 at a third of its level, and so on): a complex modulation that would otherwise need a whole patch. In the AUDIO range it is simply the six summed and gently limited.

PATCH IDEAS

1. Self-playing chords

AUDIO range, MODE on PLUCK, SPREAD fully right. A clock into TRIG 6, CHANCE around 60 %. MIX into a delay and a reverb: each beat plucks a different handful of chord notes.

2. Polyrhythmic LFOs

CV range, LOOP, SPREAD a little right of centre. Six LFOs drift in and out of phase; send them to filters, pans and levels.

3. Cascading envelopes

CV range, ONE-SHOT, SPREAD to the left. One trigger into TRIG 6 gives six envelopes of growing length.

4. Six slews from one random line

CV range, SUSTAIN, a steady +10 V into TRIG 6 and a stepped random voltage into the TWEAK input: six smoothed versions of the same line, from sharp to lazy. Quantize a few of them for harmonies that arrive at different times.

5. FM bass

AUDIO range, FM mode. A sequence into V/OCT, an envelope into the FM input, use output 1.

SPECIFICATIONS

Width	18 HP
CV range	Slope 1 from about 0.008 Hz to 128 Hz (up to ×6 with SPREAD)
AUDIO range	Slope 1 from about 12 Hz to 6 kHz, 1 V/oct
Levels	Outputs 0 to 8 V (CV) or ±5 V (AUDIO). Trigger threshold about 1 V. POLY: 6 channels.

ODETTE

Stereo repeater, from plucked strings to forty-second loops

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Odette repeats sound. SIZE picks a range of repeat times, from a couple of milliseconds, where repeats turn into strings and flangers, to forty seconds, where they become loops. TIME moves smoothly inside that range and bends the pitch like tape.

Everything is written into one long memory, so jumping from a small size to a large one brings back what was played earlier. TONE and BLUR shape the repeats a little more on every pass.

AT A GLANCE

- 8 nested sizes, from 1.3 ms to 42 s
- Smooth, pitch-bending TIME; clean jumps between sizes
- Clock sync with musical ratios, reverse, non-destructive freeze
- Stereo SPREAD and BOUNCE, tape-like DRIFT, and DUCK for live input
- 18 HP

CONTROLS

Screen	The eight sizes as a strip, the current one lit, and the current repeat time (or the clock ratio when CLOCK is patched).
SIZE	Range of repeat times; see the size table. Changing size jumps straight to the new time without bending the pitch.
TIME	Time between repeats within the size; to the right, longer. Turning it while repeats are sounding bends their pitch, like tape. With CLOCK patched it picks a ratio instead ($\times 2$ fully right, $1/1$ at the centre, $/2$ fully left) and changes jump cleanly.
FEEDBACK	Number of repeats, from one to endless and slightly beyond; saturation keeps it in check. While frozen, it chooses where in the memory the frozen loop starts.
TONE	Character of the repeats, applied again on every pass. Left: dark and warm, like an oil-can or bucket-brigade echo. Around three o'clock: neutral (default). Right: thin and crisp.
BLUR	Smears the repeats across the stereo field. Past half way, the smear is fed back into the loop and builds up over time.
SPREAD	Left and right repeat at different times, up to an octave apart in each direction.
MIX	Balance between the dry sound and the repeats. With MIX CV patched, the knob scales the CV.
DRIFT	A slow random wobble of the repeat time, different on each side, like tape.
DUCK	Lowers the repeats while sound is coming in and brings them back when it stops (fast attack, 250 ms release). Keeps a voice or an instrument clear.
SIZE · TIME · TONE	The three small knobs: amount and direction of the matching CV inputs.
REVERSE	Lit: repeats play backwards. In the smallest sizes, reversing tiny slices becomes a buzzy distortion.
FREEZE	Lit: nothing new comes in and the memory is left untouched. SIZE, TIME, FEEDBACK (as start point), TONE and BLUR keep acting on what you hear, without changing what is stored.
BOUNCE	Lit: the feedback crosses sides, so the repeats alternate left and right (ping-pong).

SIZES

Size	Repeat time	Typical use
1	1.3 to 20 ms	Plucked strings (with BEND), flanger
2	20 to 82 ms	Chorus, slapback, doubling
3	82 to 327 ms	Short echo
4	163 to 653 ms	Echo (default)
5	0.33 to 1.3 s	Long echo
6	0.65 to 2.6 s	Phrases
7	1.3 to 5.2 s	Phrases
8	2.6 to 42 s	Loops

INPUTS

SIZE · TIME · TONE	Scaled by their small knobs; 5 V covers the whole range.
FEEDBACK · BLUR	Added to their knobs; 5 V covers the whole range.
MIX	0 to 8 V, scaled by the MIX knob.
REVERSE · FREEZE	Each gate turns the matching button on or off.
CLOCK	Locks the repeats to a clock; TIME picks the ratio (see Clock sync).
BEND	Bends the repeat time. In size 1 it follows 1 V/oct, to play the repeats as a string. In other sizes it adds about 2 ms per volt, for chorus and vibrato, inverted on the right when SPREAD is off centre.
IN L · IN R	Audio inputs. IN R copies IN L when it has no cable.

OUTPUTS

OUT L · OUT R	Stereo output, ± 5 V.
PULSE	A trigger on every repeat, left or right. The light next to TIME flashes with it.

HOW IT WORKS

One memory for all sizes

Odette writes everything into one 64-second memory per side. The sizes are windows of different lengths onto that memory, so a short repeat played in size 2 is still there when you move to size 8, and repeats from smaller sizes leave traces in the larger ones.

Clock sync

With a clock in CLOCK (0.25 to 50 Hz), Odette measures its period and places the 1/1 ratio in the size that suits it best. TIME then steps through thirteen ratios from /2 to $\times 2$, and other sizes multiply or divide by powers of two. Changes jump in time without any pitch bend.

PATCH IDEAS

1. Tape echo

SIZE 4, FEEDBACK 60 %, TONE around eleven o'clock, DRIFT 30 %, BLUR 20 %.

2. Plucked string

SIZE 1, TIME fully left, FEEDBACK 85 %, MIX at the centre. Send a short noise burst or an Ernest hit into IN, and a pitch sequence into BEND.

3. Stereo chorus

SIZE 2, SPREAD a little off centre, FEEDBACK 30 %, a slow triangle LFO, turned down, into BEND.

4. Freeze and explore

Play into small sizes for a while, press FREEZE, move to size 8, then sweep FEEDBACK to travel through what was recorded.

5. A voice on stage

Microphone into IN, SIZE 5, FEEDBACK 50 %, DUCK 70 %: the echoes stay out of the way while the voice speaks and bloom in the pauses.

6. Back and forth

PULSE into REVERSE, sizes 3 to 5: the repeats turn round on every repeat.

SPECIFICATIONS

Width	18 HP
Repeat times	1.3 ms to 42 s in 8 sizes
Memory	64 s per side
Clock	0.25 to 50 Hz
Levels	Audio ± 5 V. PULSE 10 V, 1 ms. REVERSE, FREEZE and CLOCK react above about 2.5 V.